

What is running, what is broken, and what is being counted wrongly

A full audit of the 40rm IQ engine, its operations board and the registers behind it, taken on 30 September 2026. It records one defect that put a false picture of a registered dealer in front of a consumer, three faults that make a healthy system read as a broken one, and the controls that exist and are not being exercised.

WORST OUTCOME

A registered dealer read as unconfirmed

Three separate faults combined to tell a consumer a registered Canadian firm could not be confirmed on any register, beside a foreign warning about a different company.

WORST EXPOSURE

Domain lookups are down

Every RDAP attempt in the window failed. The age of a domain is the single most predictive cheap signal this product has, and no run is getting it.

MOST UNDERRATED

Nothing verifies the log on a schedule

The chain walk exists, is correct, and runs only when somebody asks for it by hand. The control is real and it is not being exercised.

GATE

The public site still says test build

The banner is accurate and it is the correct state. Clearing it needs SR-001 revised and the eight unsigned control documents signed.

01 The defect that matters most

A consumer checking a Canadian crypto exchange was shown a result headed **we could not check enough to give you a reliable answer**, carrying the sentence **we could not confirm this business on a relevant register**, with a foreign regulator's warning at the top of the record card and a single one-star review below it.

The company is registered. The Ontario Securities Commission publishes, on its own Crypto Businesses page, the crypto asset trading platforms that have received exemptive relief to offer crypto products in Ontario, with each platform's registration category and the date of its most recent decision.

Three independent faults produced that page. Each one is a defect on its own and each one is now closed or recorded.

FAULT	WHAT IT DID	STATE
The register was not in the catalogue	A register that is not in the catalogue is never asked. A register that is never asked returns nothing. Nothing was being reported to a consumer as an absence of registration rather than as a gap in our own reading.	CLOSED OSC_CATP added, link_out_only
A warning about a different company attached	The FCA warning names newton-corp.com and identifies it as a clone of Newton Investment Management Limited, firm reference 119331, at 160 Queen Victoria Street, London. It has no connection to the Toronto exchange at newton.co beyond six letters.	ENFORCED the attachment rule now also gates the time window
One review carried the weight of a regulator	A single one-star review was set out in the record card in the same shape and at the same size as a regulator's entry.	CHANGED the favourable side is now collected and both sides are ranked

The underlying error is one error, and it is worth naming plainly, because it is the error this company exists to prevent and we committed it: **an absence in our own reading was published as a finding about somebody else.**

02 What is running

Taken from the operations log over the window, not from anybody's recollection.

PART	STATE	EVIDENCE
The reasoning step	RUNNING	Written assessments are present on completed runs, and the assessment step is not named among the faults.
Search	RUNNING	351 of 511 asks answered in the window.
The database	RUNNING	108 checks recorded, 108 chain rows written. One row per completed run, which is the contract.

PART	STATE	EVIDENCE
Domain lookups	DOWN	Attempted and zero successful. Section 03.
Connector health	READS WRONG	Healthy sweeps report as refused. Section 04.
Spend	NOT RECORDED	\$0.000 across 108 checks, which is not what those checks cost. Section 04.
Chain verification	NEVER SCHEDULED	Correct, and only ever run by hand. Section 05.

03 Broken: the domain lookups

The health rule is deliberately narrow, and it is worth reading before the finding, because it is the reason this one is a real fault and not a display artefact.

```
function health(asked, answered){ if(!asked) return 'warn'; return answered>0 ? 'ok' : 'bad'; }
```

A lamp goes red only where the thing was tried and every attempt failed. Never on silence. The domain lamp is red, so RDAP was attempted on every applicable run in the window and returned nothing on all of them.

What that costs, specifically: the age of a domain is the cheapest strong signal in this product. A company claiming twelve years of operation on a domain registered six months ago is the finding that category 10 exists to produce, and category 10 cannot produce it without this lookup. Every run in the window has been running without it and no run said so on its face.

The fix, in two parts

First, restore the lookup. Second, and this is the part that outlives the outage: **a run whose strongest check did not run must say so on the result.** A degraded run that reads like a complete one is the same class of error as section 01, which is a gap in our reading reaching a consumer as a fact about a business.

04 Counted wrongly, and not counted at all

Every connector reads as refused on a healthy sweep

```
const connectorHealth = () => {
  ids.forEach(id => { const x = seen[id]; if(!x) return;
    tried += x.attempts || 0;
    ran    += (x.computed || 0) + (x.ok || 0); });
  if(!tried) return 'warn';
  return ran > 0 ? 'ok' : 'bad';
};
```

The reader adds **computed** to **ok**. The **computed** column is created by **db/telemetry.sql**, and that file has not been run against the production database. The column does not exist, the read yields nothing, **ran** stays at zero while **tried** climbs, and the rule returns bad on a sweep where nothing went wrong.

That is why every connector and all nine category tiles are red at once. A board that is entirely red is not reporting a system in trouble, it is reporting a board that cannot see. This is one **psql** run and no code change.

Spend prints zero across 108 checks

Same shape, different file. **spend_run** and **spend_client** come from **db/spend.sql**, which has also not been run. The rate variables **KBYS_RATE_CLAUDE_IN_M** and **KBYS_RATE_CLAUDE_OUT_M** are unset in the environment, so even with the tables present every row would price at nothing.

A cost control that reports zero is not a conservative cost control. It is an absent one, and the budget ceiling it is supposed to defend has never been tested against a real number.

05 The control that exists and is not exercised

The operations log is chained: every row carries the hash of the row before it and its own hash over that plus its content. Change any row after the fact and every hash after it stops matching. The walk that checks this is **verifyChain**, it is correct, and it has exactly one caller.

```
if (req.query.run === '1') { ... verifyChain(limit || null) }
```

Nothing schedules it. The gauge that reads **Runs we can prove** is not measuring the integrity of the log, it is measuring how much of the log somebody last walked by hand, which is why it reads 25 per cent. Three runs in four write a proof row and are never verified, because no verification has been run over them.

The number is not wrong. It is answering a different question from the one the label asks, which is the more dangerous of the two failures, because it looks like an answer.

The fix. A nightly walk through **api/_scheduled.js** behind **KBYS_SCHEDULE_SECRET**, an alert on any break, and a gauge whose label says what it measures. Competition Act s.74.01(1)(b) puts the onus of substantiating a performance claim on the party making it, and a chain head plus a scheduled verification is what discharging that looks like.

06 What shipped today, and why

Three changes went in against the section 01 defect and against the broader failure it is an instance of, which is that the engine only ever went looking for what was wrong.

The favourable side of the record, ranked by how hard it is to buy

RANK	WHAT GOES IN IT	MAY IT MOVE THE RESULT?
1 · Cannot be bought	A live registration, a licence in good standing, a regulator's own record, a filed financial statement, a court record showing a matter resolved, an exchange listing.	No
2 · Expensive to fake	Continuity rather than content: the same legal name across three registers, on the register since 2018, a domain eight years old, a filing history without gaps.	No
3 · Can be bought	Star ratings, review counts, testimonials, award badges, press releases.	No

The third column is the whole design. **Favourable evidence never changes the verdict.** Register standing and adverse findings set the result and everything else is context beside it. The standing block is computed after the verdict is settled, it is never read back, and there is deliberately no code path from it to the verdict.

That is the defence against purchased reviews, and it is cheaper and more reliable than any detector: once a bought five-star page has nowhere to move the needle, buying one stops working. It also has no false positives, which a detector would, and a false positive here means calling a real customer a liar.

In 2024, of the 4.5 million detected fake reviews we removed from the platform, 90% were automatically detected by our fake review detection models.

Trustpilot Trust Report 2025. 4.5 million removals represent 7.4 per cent of all reviews submitted in 2024, against 6.1 per cent in 2023. <https://corporate.trustpilot.com/trust/trust-report-2025> · read 30 September 2026

The final rule addresses reviews and testimonials that misrepresent that they are by someone who does not exist ... The final rule prohibits certain reviews and testimonials written by company insiders that fail to clearly and conspicuously disclose the giver's material connection to the business.

Federal Trade Commission, final rule banning fake reviews and testimonials, 16 CFR Part 465, published 14 August 2024 and effective 60 days later. <https://www.ftc.gov/news-events/news/press-releases/2024/08/federal-trade-commission-announces-final-rule-banning-fake-reviews-testimonials> · read 30 September 2026

The time window

Businesses make mistakes and then fix them. A fault from four years ago that was put right should not sit over a company for the rest of its life.

BAND	AGE	WHAT IT MAY DO
current	Inside 12 months	Drives the result

BAND	AGE	WHAT IT MAY DO
older	12 to 36 months	Shown and labelled. Drives nothing
archive	Beyond 36 months	Behind one control the reader can open
undated	No date on the record	Never a finding. Goes to what we could not confirm

Standing never ages. A register entry, a licence, a regulator action, a sanction and a court record describe what is true now until the issuing body changes them. A licence granted in 2019 is current. An enforcement order from 2021 that was never lifted is the present state of that party's record, not history. Ageing those out would be the single most dangerous thing this module could do, so they are exempt by class and can never fall into a band.

Why the window is arithmetic and not a judgement

“fresh” passages are consistently promoted, shifting the Top-10's mean publication year forward by up to 4.78 years ... Although larger models attenuate the effect, none eliminate it.

Do Large Language Models Favor Recent Content? A Study on Recency Bias in LLM-Based Reranking, arXiv:2509.11353. Single passages moved as many as 95 ranks, and pairwise preferences between equally relevant passages flipped by up to 25 per cent after date injection.
<https://arxiv.org/html/2509.11353v1> · read 30 September 2026

So the model carries the date and does not weigh it. The band is computed in code from the date the record itself holds, where it is arithmetic and does not have that failure mode.

The change register

Every main change to this product and to the documents that govern it now has an entry, a reason, and a report behind it, at **docs/change-register.json** and in the control panel. An entry with no reason is rejected by the build. Nothing in it may carry an identifier anybody searched, a party a check was about, or a result a check returned, and that is enforced on the write path and checked again by the build.

07 SR-001, and the registers nobody has cleared

The source register workbook was generated on 1 September 2026. It clears 175 names. The catalogue now holds 135 registers. Those two lists do not agree, in both directions.

COUNT	WHAT IT MEANS
55	Registers the engine asks on every applicable run that SR-001 does not clear. Every run counts them out of scope, so the coverage figure a consumer reads is lower than the work actually done.
95	Names the workbook clears that the catalogue has no row for. Nothing asks them, so clearing them buys nothing.
17	Registers that are published and enabled in the catalogue and waiting on a signature. Nothing asks them at all.

None of the 55 is blocked. The run asks them and the log records them as out of scope. What they need is a revision of the workbook, which is a signature and not a commit.

The 55, by category

01 · Identity and existence · 4

Florida Sunbiz · OpenCorporates · Offshore registries: Bermuda, Jersey, Guernsey, Isle of Man, Malta, Cyprus, Panama · Turks and Caicos, and Eastern Caribbean company registries

02 · Registration and licensing · 12

OSC Registered Crypto Asset Trading Platforms · SEC Form D · SFC Public Register · ESMA MiCA CASP · BC Financial Services Authority · Real Estate Council of Alberta · Autorite des marches financiers · Financial and Consumer Affairs Authority of Saskatchewan · Manitoba Securities Commission · Financial and Consumer Services Commission of New Brunswick · Nova Scotia Securities Commission · Prince Edward Island and Newfoundland and Labrador regulators

03 · Regulatory and enforcement · 8

Competition Bureau Canada · Federal Trade Commission · Consumer Financial Protection Bureau · Internet Crime Complaint Center · DOJ Press · IRS-CI · SEC Trading Suspensions · ASIC Investor Alerts

04 · People and ownership · 2

UK PSC Register · SEDI Insider Reports

05 · Legal and financial · 6

Provincial court decisions · Capital Markets Tribunal · Competition Tribunal · Securities tribunals outside Ontario · Justia Dockets · SEDAR+

06 · Web and digital identity · 1

Mail Config

08 · Transaction and payment · 3

Solscan · Tronscan · Bitcoin Explorer

09 · 4orm intelligence · 5

Analytics and Pixel Reuse · Site Technology · Host and Certificate Graph · People Cluster · Operator Graph

10 · Claim dates against the record · 14

Public Profile · Form D First Filing · Form D First Sale · EDGAR First Filing · SEDAR+ First Filing · SEDI First Insider Record · IAPD First Registration · App Store First Release · Google Play First Release · GitHub Repository Created · GitHub First Commit · First Press Release · First YouTube Video · First Social Post

The 17 waiting on a signature

GLEIF, the Global LEI Index · Cayman Islands Monetary Authority · BVI Financial Services Commission · Securities Commission of The Bahamas · Financial Services Commission of Barbados · Financial Services

08 **Monitoring the nuance of the faults**

The board today answers one question per lamp, which is whether the thing is working. That is the right question and it is not enough: every fault in this document was visible as a red lamp for some time before anybody could say what had actually failed, and two of them were not faults in the system at all.

CHANGE	WHAT IT ANSWERS
A reason on every failure	Surface the recorded reason per failed attempt from the fault log, so a red lamp opens onto what the source actually returned rather than onto a count.
Worst offenders per lamp	Which sources under this lamp are failing, ranked, so one dead connector stops presenting as nine dead categories.
Separate cannot see from is broken	A missing column, a missing environment variable and a refused request are three different states and the board paints all three the same colour. Sections 03 and 04 are two of each and they took a full audit to tell apart.
Alert on three conditions only	A lamp that was green and is now bad. A chain break. A budget ceiling reached. Anything more and the alerts get muted, which is worse than not having them.

09 **What could not be confirmed**

This section is mandatory in a control document and it is not padding. Silence reads as verified.

ITEM	WHY NOT
The live RDAP failure reason	rdap.org returned 403 to this session. The lamp establishes that every attempt failed. It does not establish why, and no workaround was attempted.
The live values behind the board	/api/stats is disallowed to automated fetching by our own robots file. The figures in section 02 are read from the board and the log, not fetched independently.
The full OSC decision for Newton	The OSC order PDF at ord_20260311_newton-crypto.pdf returned 403. The registration entry quoted in section 01 is read from the OSC's own Crypto Businesses page, which is the controlling list.

ITEM	WHY NOT
Trustpilot's weighting formula	Not published. Their public statement is that the score is weighted to reflect that newer reviews tend to be more relevant than older ones, with no thresholds given. We did not model it and we do not rely on it.
How much evidence arrives undated	The window only works where the source carries a date. Undated records are now named as gaps rather than counted, and the size of that population has not been measured on live traffic.
Terms 1.1	Counsel has not reviewed it.
tools/protocheck	Cannot run in this environment: it loads a prototype file that is not in the repository. An environment gap, not a regression.

10 The order of work

ORDER	WHAT	OWNER
1 P0	Run db/telemetry.sql and db/spend.sql against the production database. This clears the all-red board and the zero spend in one step and changes no code.	Chad
2 P0	Restore the domain lookups, and make a run whose strongest check did not run say so on the result.	Engineering
3 P1	Set KBYS_RATE_CLAUDE_IN_M, KBYS_RATE_CLAUDE_OUT_M and KBYS_SCHEDULE_SECRET in the environment.	Chad
4 P1	Schedule the chain walk nightly, alert on a break, and relabel the gauge to say what it measures.	Engineering
5 P1	Write to the Ontario Securities Commission and the Alberta Securities Commission for permission to reproduce their lists. One letter each, and it is the difference between pointing at a register and reading it.	Chad
6 P1	Revise SR-001: clear the 55, retire the 95, decide the 17, regenerate api/_sr001.json, and add SR-001 to the registry drawer so it opens like the other eleven.	Chad, with counsel

ORDER		WHAT	OWNER
7	P2	Build the fault nuance monitoring in section 08.	Engineering
8	P2	Sign the eight outstanding control documents and put terms 1.1 to counsel. These are the two gates on clearing the test build banner.	Chad, with counsel
4orm Finance · Control Standard · Internal			AUD-002 · v1.0 · 30 September 2026